

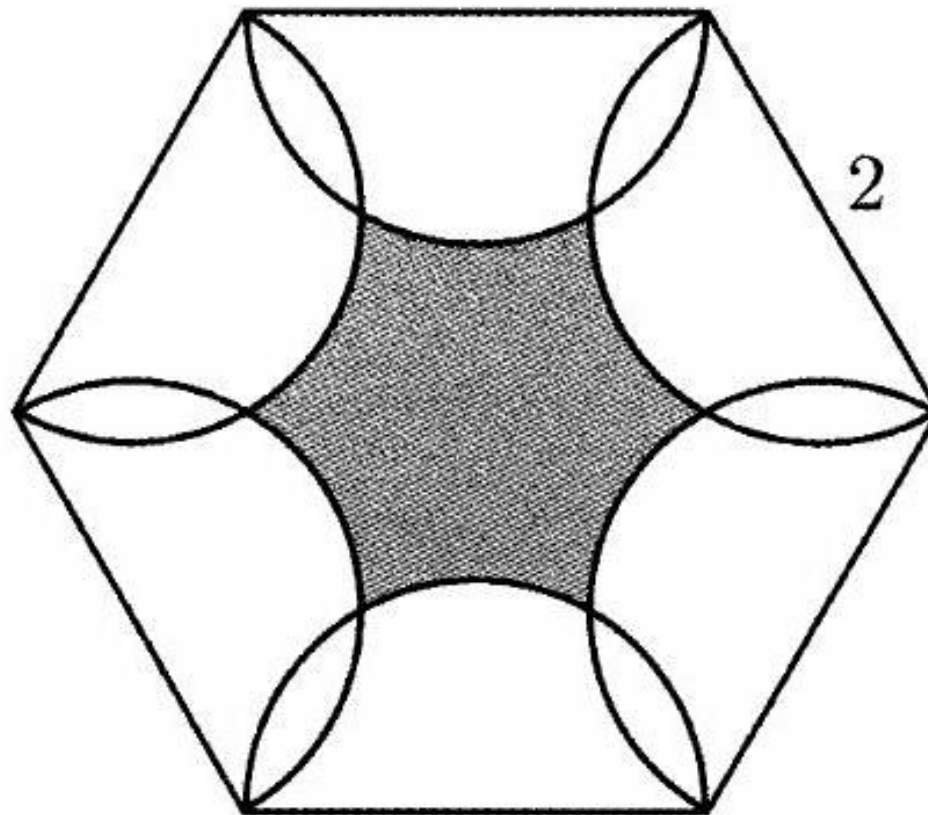
2020 AMC 10B Problem 14 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10B Problem 14. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10B Problem 14

Problem:

As shown in the figure below, six semicircles lie in the interior of a regular hexagon with side length 2 so that the diameters of the semicircles coincide with the sides of the hexagon. What is the area of the shaded region-inside the hexagon but outside all of the semicircles?



Answer Choices:

A. $6\sqrt{3} - 3\pi$

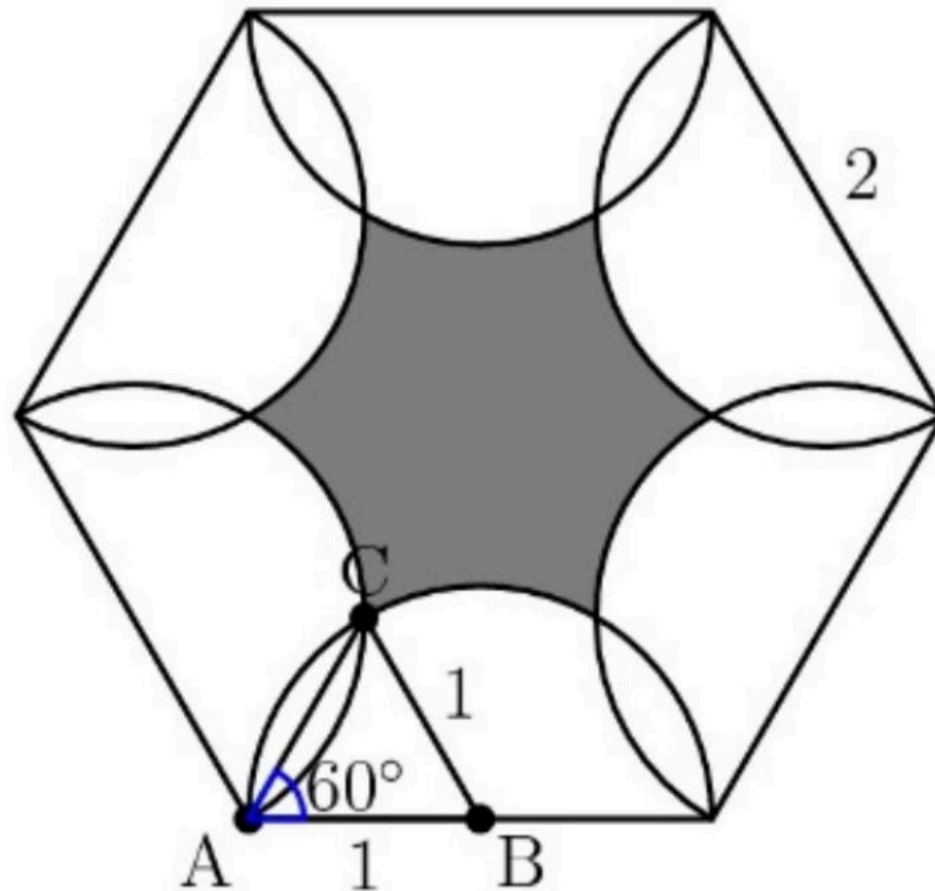
B. $\frac{9\sqrt{3}}{2} - 2\pi$

C. $\frac{3\sqrt{3}}{2} - \frac{\pi}{3}$

D. $3\sqrt{3} - \pi$

E. $\frac{9\sqrt{3}}{2} - \pi$

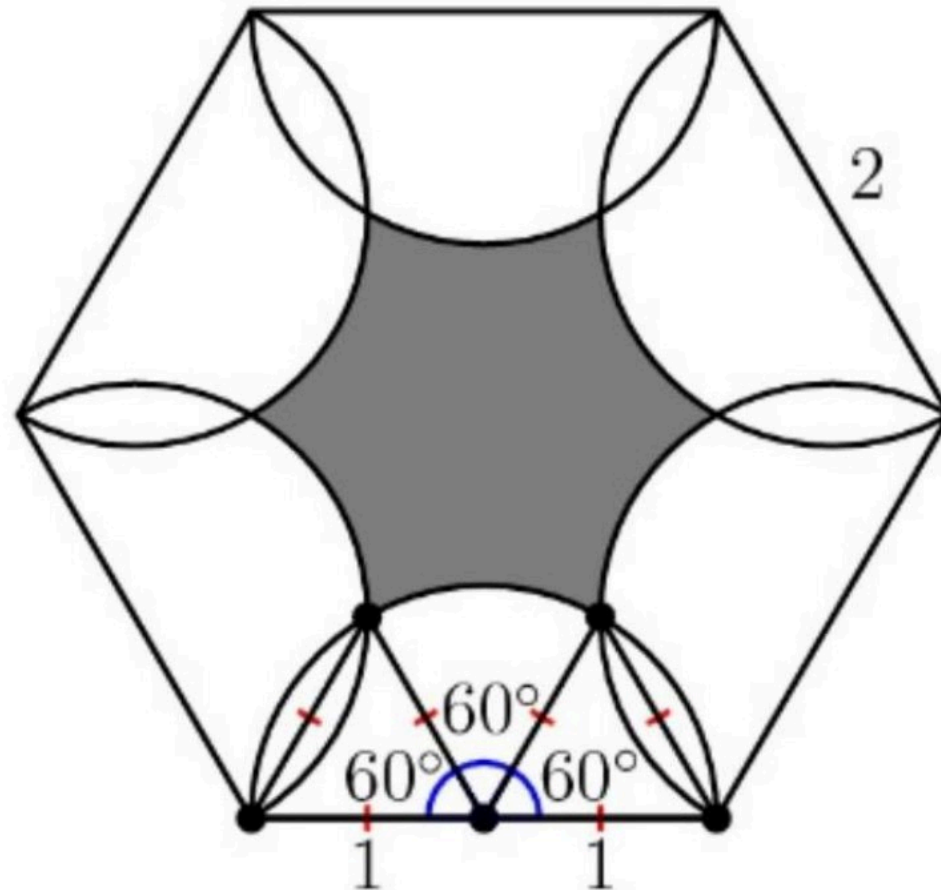
Solution:



Let point A be a vertex of the regular hexagon, let point B be the midpoint of the line connecting point A and a neighboring vertex, and let point C be the second intersection of the two semicircles that pass through point A . Then, $BC = 1$, since B is the center of the semicircle with radius 1 that C lies on, $AB = 1$, since B is the center of the semicircle with radius 1 that A lies on, and $\angle BAC = 60^\circ$, as a regular hexagon has angles of 120° , and $\angle BAC$ is half of any angle in this hexagon. Now, using the Law of Sines,

$$\frac{1}{\sin \angle ACB} = \frac{1}{\sin 60^\circ}$$

so $\angle ACB = 60^\circ$. Since the angles in a triangle sum to 180° , $\angle ABC$ is also 60° . Therefore, $\triangle ABC$ is an equilateral triangle with side lengths of 1.



Since the area of a regular hexagon can be found with the formula $\frac{3\sqrt{3}s^2}{2}$, where s is the side length of the hexagon, the area of this hexagon is

$$\frac{3\sqrt{3}(2^2)}{2} = 6\sqrt{3}$$

Since the area of an equilateral triangle can be found with the formula $\frac{\sqrt{3}}{4}s^2$, where s is the side length of the equilateral triangle, the area of an equilateral triangle with side lengths of 1 is

$$\frac{\sqrt{3}}{4} (1^2) = \frac{\sqrt{3}}{4}$$

Since the area of a circle can be found with the formula πr^2 , the area of a sixth of a circle with radius 1 is

$$\frac{\pi (1^2)}{6} = \frac{\pi}{6}$$

In each sixth of the hexagon, there are two equilateral triangles colored white, each with an area of $\frac{\sqrt{3}}{4}$, and one-sixth of a circle with radius 1 colored white, with an area of $\frac{\pi}{6}$. The rest of the sixth is colored gray. Therefore:

the total area that is colored white in each sixth of the hexagon is $2 \left(\frac{\sqrt{3}}{4} \right) + \frac{\pi}{6}$, which equals $\frac{\sqrt{3}}{2} + \frac{\pi}{6}$, and

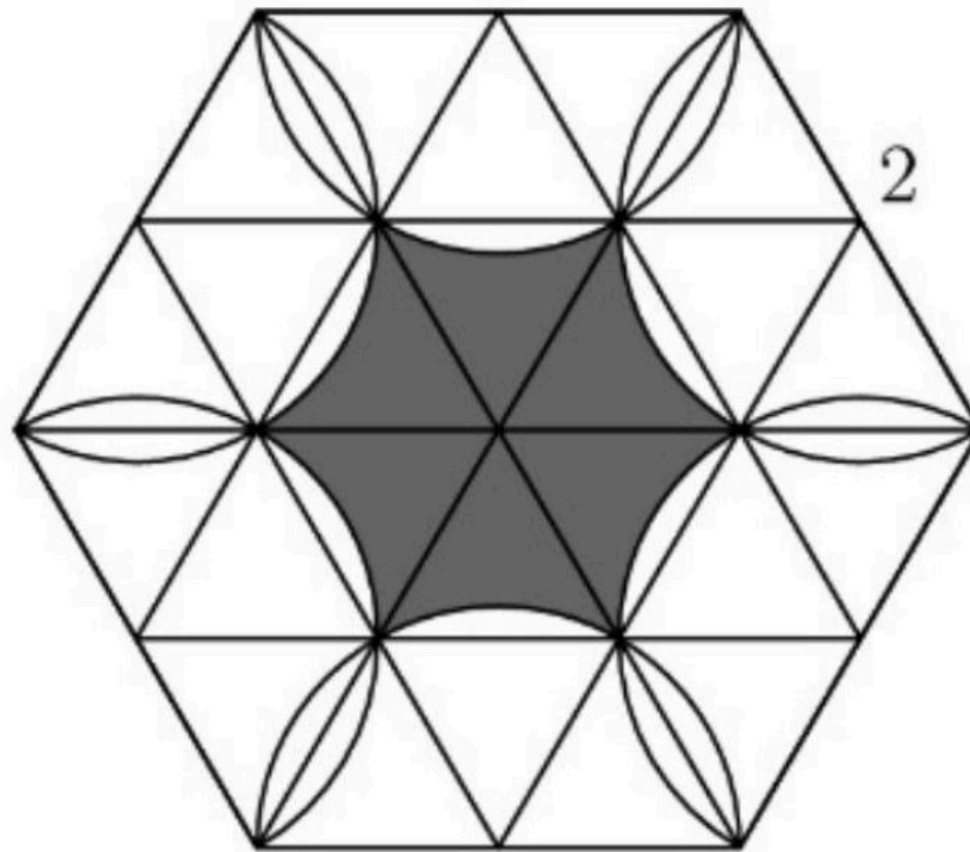
the total area colored white is $6 \left(\frac{\sqrt{3}}{2} + \frac{\pi}{6} \right)$, which equals $3\sqrt{3} + \pi$.

Since the area colored gray equals the total area of the hexagon minus the area colored white, the area colored gray is

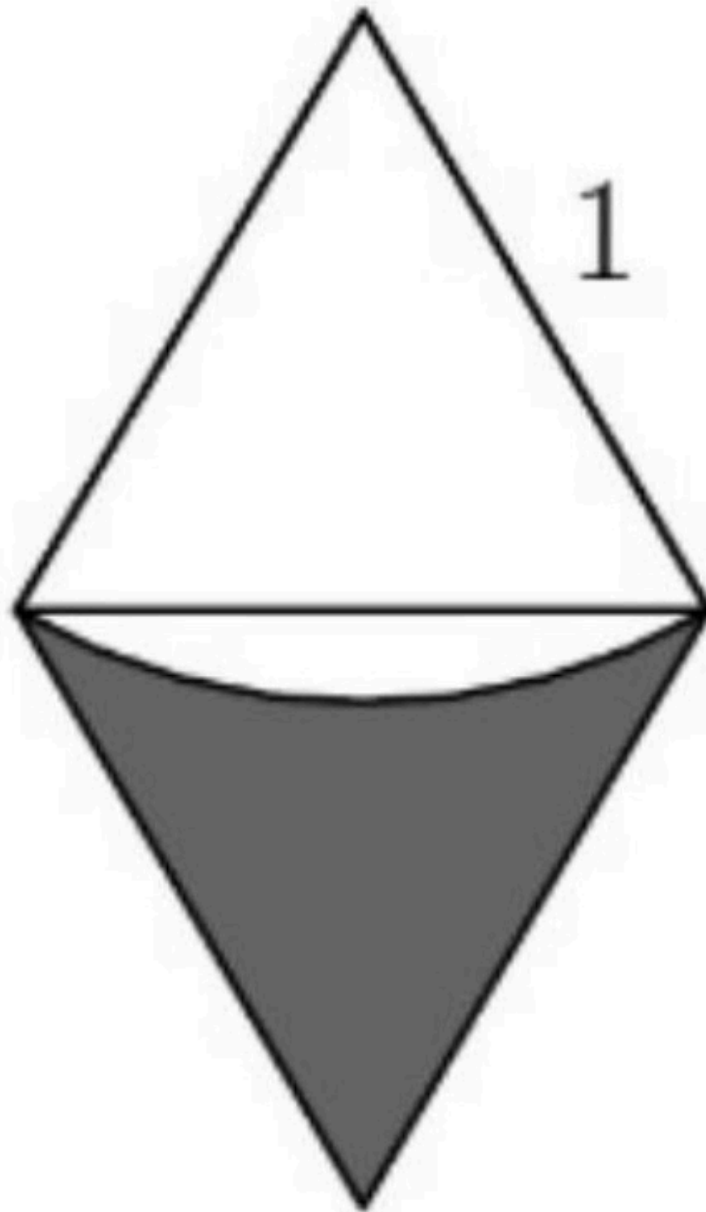
$$6\sqrt{3} - (3\sqrt{3} + \pi), \text{ which equals (D) } \boxed{3\sqrt{3} - \pi}$$

OR

First, subdivide the hexagon into 24 equilateral triangles with side length 1:



Now note that the entire shaded region is just 6 times this part:



The entire rhombus is just 2 equilateral triangles with side lengths of 1, so it has an area of:

$$2 \cdot \frac{\sqrt{3}}{4} = \frac{\sqrt{3}}{2}$$

The arc that is not included has an area of:

$$\frac{1}{6} \cdot \pi \cdot 1^2 = \frac{\pi}{6}$$

Hence, the area of the shaded region in that section is

$$\frac{\sqrt{3}}{2} - \frac{\pi}{6}$$

For a final area of:

$$6 \left(\frac{\sqrt{3}}{2} - \frac{\pi}{6} \right) = \text{(D)} \boxed{3\sqrt{3} - \pi}$$

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